Meet Udeshi

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Ph.D. Candidate, NYU Tandon School of Engineering — Expected graduation: May 2027

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SUMMARY

Cybersecurity researcher who develops benchmarks, tooling, and harnesses for LLM security agents (NeurIPS, ICML, AAAI) and builds security automation for cyber-physical systems and embedded systems (TCPS, TIST, TNSM, EMSOFT), with experience training Transformers for neural decompilation and three years of ML systems engineering at Qualcomm.

EDUCATION

NYU Tandon School of Engineering
Ph.D., Electrical Engineering — Sep 2022–present
Control/Robotics Research Lab, Center for Cybersecurity. GPA: 3.958/4
Advisors: Prof. Farshad Khorrami and Prof. Ramesh Karri

Indian Institute of Technology Bombay (IIT Bombay)
B.Tech. and M.Tech. (Dual Degree), Electrical Engineering — July 2014–June 2019
Computer Architecture and Dependable Systems Lab. CGPA: 8.18/10
Thesis Advisor: Prof. Virendra Singh

RESEARCH PROJECTS

LLM Agents for Cybersecurity — Mar 2024–present
Ph.D. research, Center for Cybersecurity, NYU

- Developed EnIGMA (ICML 2025), whose Interactive Agent Tools let LLM agents drive interactive command-line utilities such as debuggers; state of the art on NYU CTF Bench and InterCode-CTF over 350+ challenges from three benchmarks.
- Curated NYU CTF Bench (NeurIPS 2024), 200 CTF challenges for evaluating autonomous LLM security agents, with challenges and evaluations hosted in Docker; evaluated five state-of-the-art LLMs.
- Developed D-CIPHER, a multi-agent framework of collaborating planner and executor agents, with state-of-the-art solve rates of 22% on NYU CTF Bench and 20% on Cybench.
- Co-authored a lightweight CTF benchmark with LLM-as-a-judge scoring that measures how agent hyperparameters change solve rates (AAAI 2026).
- Organized the CSAW LLM Attack Challenge (2024–2026), judging the agentic solutions and agent frameworks submitted by 64 teams over three years.

Sandboxing, Digital Twins and AI Misuse for Cyber-Physical Systems — Sep 2025–present
Ph.D. research, Control/Robotics Research Lab, NYU

- Built SaMOSA, an orchestrated QEMU sandbox for malware dynamic analysis with ARM64, x86-64 and PPC64 guests and OpenPLC over Modbus TCP, which observes malware through hardware performance counters, disk, network and syscall activity (IEEE ESL / EMSOFT 2026, IOLTS 2026).
- Prototyped *Ransomware 3.0/PromptLock* to evaluate LLM-powered ransomware lifecycles with Lua code generation and stage-wise attack chain execution. Covered by NYU Tandon News (2025) and MIT Technology Review (2026).
- Designed an LLM binary-diffing framework for software supply chain security that detects malicious code in software updates, with LLM inference and batch completion via vLLM on parallel HPC jobs.

Reverse Engineering of Cyber-Physical Systems — Aug 2022–present
Ph.D. research, Control/Robotics Research Lab, NYU

- Trained Fairseq Transformer models (up to 12M parameters) for neural decompilation on the NYU HPC.
- The resulting REMEND decompiler recovers equations from binaries (ACM TIST). One model covers three ISAs, three optimization levels and two languages, with 89.8–92.4% accuracy on C and Fortran, 8% above state-of-the-art on real-world code, and 0.13 s per function.
- Designed REMaQE (ACM TCPS), an `angr`-based symbolic execution framework that recovers math equations from controller binaries; it is 100% accurate on C and Simulink binaries.
- REMaQE found a bug in the Linux `tmon` PID controller; submitted a kernel patch (2023).

Tamper-Proof Network Measurements on a SmartNIC
Ph.D. research, Control/Robotics Research Lab (IEEE TNSM 2025)
Designed tamper-proof traffic measurement for intrusion detection on a Netronome SmartNIC (P4, embedded C), out of reach of host-OS rootkits. Two designs (associative table, count-min sketch) run at 10 Gbps with 99.5% of baseline throughput and 12–23 µs added latency.

Hardware Security (Masters Thesis) Aug 2018–June 2019
CADSL, IIT Bombay. Advisor: Prof. Virendra Singh
Designed cache side-channel attacks that disable the hardware prefetcher to strip its noise (against ciphers) and that leak between SMT threads through the shared re-order buffer (via IPC). Built an LLVM-IR framework for x86-to-ARM program migration in the heterogeneous-ISA core HIDC.

WORK EXPERIENCE

Siemens Healthineers USA, Princeton, NJ May 2026–Aug 2026
AI Cyber Intern, AI Factory

- Contributed to ARPA-H UPGRADE to make vulnerability patching more efficient for clinical and hospital setups, and modeled the cyber-clinical impact of unpatched vulnerabilities in that critical infrastructure.
- Contributed to LLM-based generation of synthetic vulnerabilities for hospital devices and networks, as training data for an RL vulnerability-patching agent.
- Ran multi-GPU vLLM inference on single nodes, and multi-node, multi-CPU hospital simulations as dependency-linked Slurm jobs.

Qualcomm R&D, Bangalore, India July 2019–July 2022
Senior Engineer, ML Compiler Team for Cloud AI100 Accelerator

- Built core parts of the Cloud AI100 ML inference compiler: multi-core, multi-thread data tiling, memory management, graph scheduling and operator fusion, with graph optimizations for 2D/3D vision, recommendation and autonomous-driving models.
- Implemented accelerator kernels for the input and output processing stages inside ResNet and YOLO graphs. Worked deep in PyTorch and TensorFlow graph-capture and op-lowering internals; contributed to the open-source *PyTorch Glow* compiler.
- Explored polyhedral loop and array optimization alongside Halide-style scheduling. Deployed a power-efficient object tracking pipeline using Kernelized Correlation Filters (KCF) on AI100.

Amazon India Jul 2017–Aug 2017
Software Development Intern, Transportation Financial Systems
Processed and sorted 1 million+ receipts daily (DynamoDB, SQS); automated setup of 30+ AWS resources with CloudFormation.

Google Summer of Code, Kivy May 2016–Aug 2016
Python Native UI Framework Game Engine
Built a Python/Cython Tiled maps module and a Cython-optimized entity-component animation system for the KivEnt game engine.

SELECTED PUBLICATIONS

Equal contributions are marked with †.

[C1] T. Abramovich, **M. Udeshi**†, M. Shao†, K. Lieret, H. Xi, K. Milner, S. Jancheska, J. Yang, C. E. Jimenez, F. Khorrami, P. Krishnamurthy, B. Dolan-Gavitt, M. Shafique, K. Narasimhan, R. Karri, and O. Press. "EnIGMA: Interactive Tools Substantially Assist LM Agents in Finding Security Vulnerabilities." *International Conference on Machine Learning (ICML)*, 2025.

[C2] M. Shao†, S. Jancheska†, **M. Udeshi**†, B. Dolan-Gavitt†, H. Xi, K. Milner, B. Chen, M. Yin, S. Garg, P. Krishnamurthy, F. Khorrami, R. Karri, and M. Shafique. "NYU CTF Bench: A Scalable Open-Source Benchmark Dataset for Evaluating LLMs in Offensive Security." *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.

[C3] M. Shao, N. Rani, K. Milner, H. Xi, **M. Udeshi**, S. Aggarwal, V. S. C. Putrevu, S. K. Shukla, P. Krishnamurthy, F. Khorrami, R. Karri, and M. Shafique. "Towards Effective Offensive Security LLM Agents: Hyperparameter Tuning, LLM as a Judge, and a Lightweight CTF Benchmark." *AAAI Conference on Artificial Intelligence*, 40(35):29660–29668, 2026.

[C4] **M. Udeshi**, V. S. C. Putrevu, P. Krishnamurthy, R. Karri, and F. Khorrami. "An Orchestrated Sandbox for Side-Channel Analysis of Cyber-Physical Systems Malware." *IEEE Embedded Systems Letters*, presented at *Embedded Software Conference (EMSOFT)*, 2026.

[C5] **M. Udeshi**†, M. Raz†, P. Krishnamurthy, R. Karri, and F. Khorrami. "Sandbox-Enabled Digital Twin for Cyber-Physical Systems." *IEEE International Symposium on On-Line Testing and Robust System Design (IOLTS)*, 2026.

[J1] **M. Udeshi**, P. Krishnamurthy, R. Karri, and F. Khorrami. "REMEND: Neural Decompilation for Reverse Engineering Math Equations from Binary Executables." *ACM Transactions on Intelligent Systems and Technology*, 17(3), 2025.

[J2] **M. Udeshi**, P. Krishnamurthy, R. Karri, and F. Khorrami. "Tamper-Proof Network Traffic Measurements on a NIC for Intrusion Detection." *IEEE Transactions on Network and Service Management*, 22(2), 2025.

[J3] **M. Udeshi**, P. Krishnamurthy, H. Pearce, R. Karri, and F. Khorrami. "REMaQE: Reverse Engineering Math Equations from Executables." *ACM Transactions on Cyber-Physical Systems*, 8(4), 2024.

HONORS AND AWARDS

DAC Young Fellowship — Design Automation Conference, 2023

Student Travel Grant to present at ESWEEK (EMSOFT) — IEEE CEDA, 2026

RoCStar (Recognition of Outstanding Contributions), AI100 compiler and KCF — Qualcomm, 2021

Gold Medal, Indian National Physics Olympiad (top 35 nationwide) — HBCSE, 2014

SERVICE AND PEER REVIEW

Peer Review: AAAI 2026, AAAI 2027, ICLR 2026, ICML 2026, NeurIPS 2026, CPS-IoTSec 2025, *ACM Transactions on Intelligent Systems and Technology*, *The Computer Journal*, *The Journal of Supercomputing*, *Journal of Computer Virology and Hacking Techniques*.

2016–2017	Manager, Electronics Club	IIT Bombay
2015	Reviewer, 46th International Physics Olympiad	Mumbai, India
2014	Volunteer tutor, Supplemental Learning Program	Vidya NGO Mumbai

TEACHING AND MENTORING

2020–2021	Mentor, hardware security project, Qualcomm Innovation Fellowship.	Qualcomm
Spring 2019	Teaching Assistant, VLSI Design Lab (Prof. Sachin Patkar).	IIT Bombay
Fall 2018	Teaching Assistant, Microprocessors (Prof. Virendra Singh).	IIT Bombay
Summer 2016	Teaching Assistant, CS101 Computer Programming (Prof. D. B. Phatak).	IIT Bombay

OPEN SOURCE

Linux kernel (`tools/thermal` patch), Kivy (KivEnt, GSoC 2016), beets, fast_playlist.

SKILLS

Agents, LLMs	ReAct-style agents, planner–executor multi-agent systems, benchmark design, LLM-as-a-judge, LLM function calling, vLLM
ML	PyTorch, TensorFlow, Fairseq, Transformer training, ML compilers (PyTorch Glow, LLVM), polyhedral optimization
Security	Symbolic execution (`angr`), static/dynamic binary analysis (Binary Ninja, Ghidra, `radare2`), neural decompilation (Fairseq), binary diffing (`ghidriff`), malware sandboxing (QEMU), side-channel analysis (gem5, PIN)
Programming	Python, Embedded C/C++, C, C++, Shell/Bash, Assembly, JavaScript, HTML, Java
Coursework	Hardware Security & Trust, Deep Learning, Advanced Computer Architecture, Advanced Hardware Design, Computer Vision, VLSI Design